ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech (CSE) (CC)

Semester: VI

SUBJECT CODE: BCSI601

ARTIFICIAL INTELLIGENCE

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(10 Marks)**

Q. 1. Define the following terms: **(1 × 5 = 5 Marks)**

(i) What are the component of NLP?

(ii) How do intelligent agents differ from AI models?

(iii) What is computer vision?

(iv) What are some common AI applications?

(v) Give the general model of learning agent.

Q. 2. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) What is searching for Solutions?

(ii) Give two examples of uninformed search algorithms.

(iii) How can local search algorithms be made more robust?

(iv) What is adversarial search?

(v) Give some key points about Alpha-Beta pruning.

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) What is the difference between Forward Chaining vs Backward Chaining.

OR

(b) What is the difference between Propositional logic vs First Order Predicate logic.

(ii)(a) How is knowledge represented in ontological engineering, and what role does ontological engineering play in building intelligent systems?

OR

(b) Describe A* search technique and prove that it is complete and optimal.

Q.4.(i) (a) Write short note on reinforcement learning & machine learning

OR

(b) Write short note on discrete model & Naive Bayes Model

(ii)(a) What is Bayesian network? Explain steps to create a Bayesian network.

OR

(b) Distinguish between Markov model and Hidden Markov Model in probabilistic reasoning.

SECTION – C

NOTE: Attempt any two questions. (10 × 2 = 20 Marks)

Q.5.(a) Define Principle Component Analysis (PCA). Explain how PCA is used in pattern recognition. Also describe parameter estimation methods in pattern recognition.

OR

(b) What is K-Means Algorithm? How does the K-Means Algorithm Work? How to choose the value of "K number of clusters" in K- means Clustering?

Q.6.(a) Discuss support vector machine and Linear Discriminant Analysis (LDA).

.OR

(b) Show, how classification is done by K-nearest neighbors. Construct KNN classification Algorithm on the Following data set and predict the class for X (p1=4, p2=6). Given k=3

P1	P2	Class
6	5	False
7	7	False
3	5	True
2	4	True